

Speeding up network motif discovery by counting isomorphic subtrees*

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Abstract. We propose a new algorithm for counting the number of subgraphs of a network isomorphic to a given query graph, in the context of network motif search. Vertices of high degree (hubs) contribute to a combinatorial explosion in the total number of subgraphs of a network. Many real world networks tend to contain large hubs, which makes existing motif search algorithms intractable for motif sizes greater than 7 on a wide variety of networks of interest. Our procedure leverages the k -core decomposition and a novel subtree-counting procedure to quickly scan the periphery of a network, producing significant speedup in the presence of trees and peripheral hubs. Namely, we prove that $\# \text{ROOTEDSUBTREEISOMORPHISM}$ is $\# \mathbf{P}$ -complete via a reduction from counting bipartite matchings. We build on this result to propose a fast subgraph-counting algorithm, provide analytic upper bounds on its execution time, and test it on a corpus of real network data.

Key words. network motifs, subgraph isomorphism, k -core decomposition

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1. Introduction. A network is a useful abstraction for studying structural patterns and properties in many types of systems [18, 29]. One way to study such patterns is by identifying repetitions of certain subnetworks, following the reasoning that in many systems, the frequent occurrence of a “module” is an indicator of its importance in the function of the system. Patterns in a network that appear with statistically significant frequency with respect to a null model (an appropriately defined random network) are called *network motifs*. Examples of networks where motifs have been studied include class diagrams in software architecture [32], brain networks [28], transcriptional regulation networks [27] and many more [1, 10, 19, 21].

Since their introduction [17], an active line of research has been dedicated to the design of algorithms that search for such motifs ever more efficiently. Most of these algorithms fall under one of two categories: network-centric algorithms, in which all [12, 15, 17] or a sample [13, 33] of size- k subgraphs of a given network are enumerated, and motif-centric algorithms [9, 20] in which size- k graphs are enumerated *a priori* and mapped onto a network one by one. Typically in the former category, after enumeration, subgraphs are grouped into isomorphic equivalence classes and their frequencies are then calculated or estimated. Two of the advantages of motif-centric approaches is the possibility of searching for the instances of a single subgraph in a network, and the reduced cost of computing isomorphisms for each subgraph incurred in the classification task of network-centric algorithms. Additional ideas such as symmetry-breaking [9] and expansion trees [20] have enabled further speedups, making it possible to search for all subgraphs up to size 8, and individual subgraphs up to size 15 [9]. The *g-trie*, introduced in [23], is an efficient data structure that enables the search for any set of subgraphs, taking advantage of the similarity between subgraphs in that set. This minimizes

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the cost of redundant searches incurred by motif-centric approaches, which execute as many queries as there are subgraphs, disregarding their potential similarity, while eliminating the unnecessary isomorphism checks of network-centric subgraph enumeration. This method used in combination with symmetry-breaking produces a vast improvement compared with pre-existing approaches [24]. Additional techniques have included focusing only on tree motifs [15], parallelizing subgraph enumeration [26] or subgraph mapping [16].

A few remarks serve to lay out our strategy for an improved motif search algorithm. First, we note that existing approaches to motif search all operate by either enumerating or mapping *instances* of subgraphs inside a network. In principle, to discover motifs it would suffice to count all subgraphs of a network, rather than list them. The counting version of the subgraph isomorphism problem, #SUBGRAPHISOMORPHISM, asks how many subgraphs of a graph G are isomorphic to another graph H . Algorithms for #SUBGRAPHISOMORPHISM include [6] and [2]. The latter color-coding technique achieves polynomial time when counting non-induced subgraphs of bounded treewidth, provided $|H| = O(\log(|G|))$. However it was recently proved [5] that boundedness of the vertex cover number of H is the only condition that guarantees tractability for #SUBGRAPHISOMORPHISM. Regardless of asymptotic behavior, counting subgraphs is arguably faster than listing them, simply owing to the memory access and storage overhead when large numbers of instances are involved.

Another issue with existing algorithms is that none of them account for the difference in runtime incurred by different subgraph topologies. Early on in the investigation of network motifs, it was suggested [11] that certain subgraphs could contribute disproportionately more to the runtime of motif search because of combinatorial explosions in the number of times they appear in a network. A good example is the star graph of size k , of which there are $\binom{n-1}{k-1}$ instances in a star graph of size $n \geq k$.

Lastly, scale-free networks are known to possess large hierarchically structured hubs [3] which can quickly make the motif search problem intractable in the contexts where such networks show up. Putting aside the discussion of how ubiquitous scale-free networks really are [4], biological networks, where investigations of network motifs originated, often also contain prohibitively large hubs [22] whether their degree distributions are scale-free, geometric, or log-normal. As we will check experimentally, the runtime of motif search can even be dominated by a single star motif while searching among hundreds or thousands of motifs, especially when the target network contains large hubs.

Taking into account the above remarks, we make #SUBGRAPHISOMORPHISM our focus in the present paper, and take advantage of both motif and network topology to accelerate subgraph counting. We show that a special case of this problem where the network G and query H are rooted unlabeled trees is #P-complete. We then build on Grochow-Kellis [9] to design a subgraph-counting algorithm that leverages tree and hub structure in a network. Our main contribution consists in separating each of the network and the query into two subsets using the k -core decomposition: the “periphery”, which contains all the low-connectedness nodes, and the more interconnected nodes, which can be viewed as “the core”. A key feature of this decomposition is that it guarantees that the periphery consists entirely of trees. This enables us to run the peripheries of both network and query through a fast subtree-counting algorithm, saving considerable time compared with standard motif search procedures. In addition, the same decomposition offers a powerful early abort for motif search; the *coreness*

of a vertex imposes a strong constraint on the possible isomorphisms that can map a query graph to a target network, allowing our algorithm to waste less time trying out mappings that do not yield valid instances.

In [section 2](#) we make the notions of core and periphery rigorous, analyze the time complexity of subtree-counting and provide an algorithm to that end. [Section 3](#) introduces our main algorithm and provides an analytic treatment of its performance. We then test the algorithm on a corpus of real networks, and relate its empirical performance to several key network statistics in [section 4](#). We conclude in [section 5](#).

2. Counting subtrees in the 1-shell.

2.1. The k -core decomposition. The k -core decomposition [14] is a procedure that reveals the “core-periphery” structure of a given network. A k -core is defined as the maximal sub-network that contains only vertices of degree at least k . The k -shell is the set of vertices in the k -core that are not part of the $k + 1$ -core. A vertex inside the k -shell is said to be of *coreness* k . The *degeneracy* of a network is the maximal coreness of any of its vertices. The k -core decomposition can be carried out in linear time as follows: remove all vertices of degree 1 until there are none left. Those vertices constitute the 1-shell. Make additional passes removing all vertices of degree at most k at each pass, labeling each vertex with its coreness k .

In any network with degeneracy 2 or more, the 1-shell consists entirely of trees, each of whose root belongs to the 2-core. To see why, notice that no tree can contain a nontrivial subgraph where every vertex is of degree 2 or more. Performing a k -core decomposition on H and G reveals these trees, if they exist. By selecting one root at a time from each of H and G ’s 1-shell trees, we can then run a subtree-counting algorithm as a subroutine to any standard motif-centric algorithm. We build on the Grochow-Kellis symmetry-breaking algorithm [9], but in principle our subtree-counting technique can be used with any standard motif-centric algorithm. In practice however, there are subtleties of implementation that have to do with symmetry-breaking and the 1-shell/2-core boundary that should be taken care of, and will be discussed in [subsection 3.1](#). In the following we discuss the computational complexity of subtree-counting and motivate its use in the context of motif search.

2.2. Complexity of counting subtrees. Given two rooted, unlabeled trees T and S , we define $\# \text{ROOTEDSUBTREEISOMORPHISM}$ as the problem of counting the number of subtrees of S isomorphic to T . This problem is $\# \mathbf{P}$ -complete, as we prove in [Theorem 2.1](#) via a reduction from $\# \text{IMPERFECTMATCHINGS}$, the problem of counting the number of matchings in a bipartite graph. The latter problem was proved $\# \mathbf{P}$ -complete by Valiant [31]. As does Valiant, we define a counting Turing machine as a standard nondeterministic TM that can output the count of its accepting configurations on an auxiliary tape. We write $A \leq_T B$ if language A is Turing-reducible to language B , $T \preceq S$ if there exists a root-preserving isomorphism between tree T and a subtree of S , and $T \preceq! S$ if such an isomorphism is unique up to the symmetries of T .

Theorem 2.1. $\# \text{ROOTEDSUBTREEISOMORPHISM}$ is $\# \mathbf{P}$ -complete.

Proof. Let S and T be two rooted, unlabeled trees. Membership in $\# \mathbf{P}$ is easy to verify: consider a counting Turing machine that removes leaves from S nondeterministically and

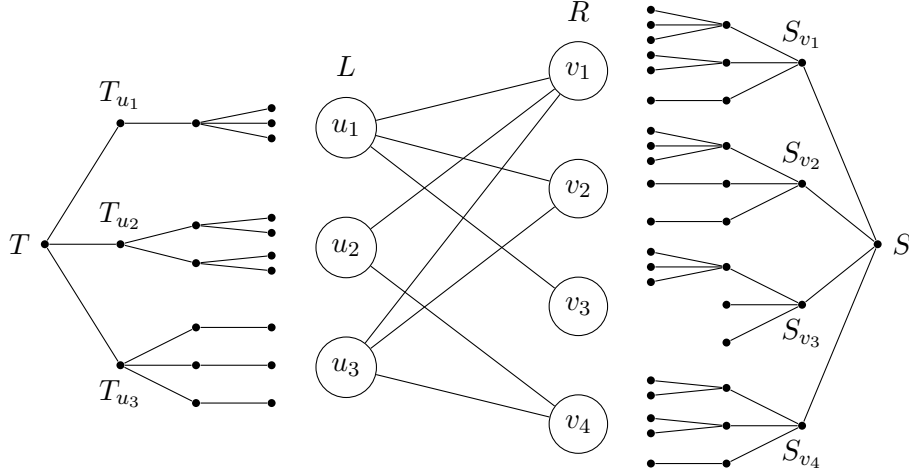


Figure 1. Graphical example of equivalence (2.1).

checks for tree isomorphism with T in linear time by simply comparing preorder traversal sequences. The number of accepting configurations is equal to the number of instances of T in S .

Let $B = (L, R, E)$ be a bipartite graph where L and R are respectively the sets of left and right vertices, and $E = \{(u, v) | u \in L, v \in R\}$ is the set of edges. To show that $\# \text{IMPERFECTMATCHINGS} \leq_T \# \text{ROOTEDSUBTREEISOMORPHISM}$, we construct a tree T_u for each vertex $u \in L$, and a tree S_v for each $v \in R$ (see Figure 1), such that

$$(2.1) \quad (u, v) \in E \iff T_u \preceq! S_v$$

On the left side of the bipartite graph, we then create a tree by adding an edge between a new vertex t and the root of T_u , for each $u \in L$. Every T_u is now a subtree of the new tree T , rooted at a child vertex of t . Similarly, we create tree S on the right side. This establishes a one-to-one correspondence between a matching of B and an instance of T in S (up to symmetry).

We construct the $\{T_u\}_{u \in L}$ trees as follows: $\forall i \in \{1, \dots, |L|\}$, T_{u_i} has i depth-1 vertices, and each of those has $|L| - i + 1$ children. For the $\{S_v\}$ trees, $\forall j \in \{1, \dots, |R|\}$, S_{v_j} has exactly $|L|$ depth-1 vertices. We then populate the children of those vertices by “inserting” one copy of T_{u_i} into S_{v_j} if and only if $(u_i, v_j) \in E$; in other words, i among S_{v_j} ’s depth-1 vertices has at least $|L| - i + 1$ children, and the remaining $|L| - i$ depth-1 vertices have no children. This construction guarantees that at most one instance of T_{v_i} appears in S_{v_j} .

Now if a Turing machine can query an oracle for $\# \text{ROOTEDSUBTREEISOMORPHISM}$ on instance (T, S) , the answer is equal to the number of bipartite matchings in graph B , thus establishing the result. ■

Like bipartite matching, this result is surprising considering that its corresponding decision problem, RootedSubtreeIsomorphism, is in $\mathbf{P}$. We note that this result is different from the one proved by Goldberg and Jerrum in 2000 [8], in which counting all rooted subtrees which are *distinct* up to isomorphism within a given tree is proved $\#\mathbf{P}$ -complete, whereas our result

concerns counting all rooted subtrees of a tree *equivalent* up to isomorphism to a given tree (disregarding the symmetries of that tree).

2.3. Description of the subtree-counting algorithm. The permanent of an $n \times n$ matrix $A = (a_{ij})$ is defined as

$$(2.2) \quad \text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}$$

where S_n is the set of all permutations of the index set. Valiant showed [30] that the number of matchings of a bipartite graph is equal to the permanent of its (0,1)-adjacency matrix. In a weighed bipartite graph B with a square adjacency matrix $M = (m_{ij})$, if we define the weight of a matching s as the product of the weights of all its edges

$$(2.3) \quad w(s) = \prod_{i=1}^n m_{i, s(i)}$$

then the permanent of M is the sum of the weights of all matchings of B . If we assign the weight of an edge (u, v) to be the number of matchings between T_u and S_v , the permanent of M becomes the total number of instances of T in S . In the case where M is rectangular with dimension $m \times n$ where $n \geq m$, $\text{per}(M)$ is the sum of the permanents of all $m \times m$ square sub-matrices of M .

If the roots of trees $\{T, S\}$ have k and n children respectively, we need to compute $\binom{n}{k}$ permanents of $(k \times k)$ matrices to count all instances of T in S . Hubs that appear in trees can make $\binom{n}{k}$ a very large number, and permanents of $k \times k$ matrices expensive to calculate. Note that these computations happen for every non-leaf vertex of T , as we elaborate below. We propose the following simplification to address this issue:

Algorithm 2.1 counts all the instances of an unlabeled tree T inside an unlabeled tree S , where the roots of both trees are given by t and s respectively. D is a dictionary that stores at key u the preorder traversal string of the subtree of T rooted at u for all nodes $u \in T$ (here we abuse the term “subtree” by having it refer to the subgraph of T that contains u and all its descendants relative to the root of T).

We are interested in counting only one copy of T inside S for each one of T ’s automorphism classes, since we are dealing with unlabeled trees where symmetries do not matter. To account for this, the algorithm starts by listing the preorder strings of the depth-1 subtrees of T , and grouping the identical strings among that list into equivalence classes (recall that a preorder traversal uniquely determines an unlabeled tree, up to symmetry). There are $n!$ possible permutations of the elements of an equivalence class of subtrees, whence the correction factor *multiplicity*.

Next, the algorithm checks for the base case, where the tree T consists of only one node with no children. Otherwise, it loops through every pair of subtrees (T_u, S_v) rooted at depth-1 and recursively counts the instances of T_u in S_v , storing the count in a matrix M at entry $[u][v]$. This is equivalent to constructing a bipartite graph and assigning a weight k to edge

Algorithm 2.1 COUNTROOTEDSUBTREES(T, S, t, s, D)

```

subtreesPreorderStrings :=  $\{D[i] \mid \forall i \in t.children\}$ 
equivalenceClassSizes :=  $\{\text{countStringsEqualTo}[k] \mid \forall k \in \text{subtreesPreorderStrings}\}$ 
multiplicity := 1
for  $n$  in equivalenceClassSizes do
    multiplicity := multiplicity  $\times n!$ 
end for
if  $t.children = \emptyset$  then
    return 1
end if
for  $u$  in  $t.children$  do
    for  $v$  in  $s.children$  do
         $T_u := \text{detachTree}(T, u, t)$  //the “downward-facing” subtree of  $T$  rooted at  $u$ 
         $S_v := \text{detachTree}(S, v, s)$ 
         $k := \text{COUNTROOTEDSUBTREES}(T_u, S_v, u, v, D)$ 
         $M[u][v] := k$ 
    end for
end for
return  $\text{permanent}(M) \div \text{multiplicity}$ 

```

187 (u, v) . Finally, the permanent of M is computed using Ryser’s formula:

188 (2.4)
$$\text{perm}(M) = \sum_{k=0}^{n-1} (-1)^k \Sigma_k$$

189 where Σ_k is obtained by summing the products of the row-sums of A_k , the matrix obtained
 190 by deleting k rows from M , for all combinations of rows. The worst-case time complexity of
 191 using Ryser’s formula for the permanent is $O(2^{n-1}n^2)$ [25]. However, to get an idea of how
 192 much faster **Algorithm 2.1** is than regular motif search over trees, see **(add empirical speed**
 193 **comparison)**.

194 3. The subgraph-counting algorithm.

195 **3.1. Boundary conditions.** Recall from the introduction that our method consists in sepa-
 196 rating the 1-shell from the 2-core and performing subtree-counting on the former, then running
 197 a regular motif search algorithm on the latter. In doing so we must take care of a few boundary
 198 cases that arise at the interface between the 1-shell and the 2-core **(add figure to visualize**
 199 **1-shell and 2-core)**. To see why, let $H = (C_H, P_H)$ be a query graph and $G = (C_G, P_G)$
 200 be a network, where C and P refer respectively to the 2-core and the 1-shell. An instance
 201 of H could appear “halfway” between the 1-shell and the 2-core of G such that only part of
 202 P_H extends into P_G . Because of this, to properly count every instance of H in G , we must
 203 run each tree $T_H \in P_H$ as well as all of its subtrees through **Algorithm 2.1**. By using the
 204 symmetry-breaking technique introduced in [9] on T_H , we can cut down the cost incurred by
 205 enumerating all possible subtrees by an amount proportional to the product of the number of
 206 symmetries at each level of trees of P_H .

On the flip side, another boundary condition provides us with a powerful early abort when searching for instances of H inside G : suppose that H appears halfway between C_G and P_G , only this time C_H extends partially into P_G . As we prove below, this cannot happen. In fact, it cannot happen between a k -core in H and an h -shell in G for any integers $h < k$ without violating induced isomorphism, which we require in the formal definition of “an instance of H inside G ”.

Definition 3.1. We say that an instance of $H = (V_H, E_H)$ appears in $G = (V_G, E_G)$ when there exists an induced isomorphism $\phi : V_H \mapsto V_G$ between the nodes of H and those of G :

$$(3.1) \quad \forall u, v \in V_H, (u, v) \in E_H \iff (\phi(u), \phi(v)) \in E_G$$

Theorem 3.2. For any k , a mapping between a vertex $v \in H$ of coreness k onto a vertex $u \in G$ of coreness $h < k$ violates induced isomorphism between H and G .

Proof. First notice that a vertex v in a graph’s k -shell can have at most k neighbors in the $k + 1$ -core. Otherwise, it wouldn’t be removed when the k -core decomposition is making passes to remove vertices of degree at most k , and hence would not be part of the k -shell by definition. By the same reasoning, a vertex u in the $k + 1$ -core must have at least $k + 1$ neighbors inside the $k + 1$ -core. The above definition implies that an isomorphism that maps u to v should also map at least one of u ’s $k + 1$ -core neighbors to a neighbor of v in the k -shell. Proceeding by induction, we see that v has at least $k + 1$ neighbors, each of which has at least $k + 1$ neighbors, and so on. Therefore v must be in the $k + 1$ -core, a contradiction. ■

We can use this fact to create an early abort by coreness: our algorithm should never bother to try to map a vertex in H to another in G with lesser coreness.

3.2. Main result. We build on the Grochow-Kellis symmetry-breaking algorithm, and refer to the function from their paper by the same name. **Algorithm 3.1** takes in two graphs H and G , and returns the number of images of H that appear in G . It starts by running a k -core decomposition on both graphs. Then, as in the Grochow-Kellis algorithm, it finds all automorphisms from H onto itself by calling `FINDSUBGRAPHINSTANCES( $H, H$ )` [9], groups the nodes that map to each other in the same equivalence class and returns a representative node from each class. We abbreviate this behavior in the function `findEquivalenceClassRepresentatives`. The symmetry-breaking conditions are also computed similarly as in [9].

The next part of the algorithm consists in listing all the subtrees of the 1-shell of H , up to symmetry. This is done by the function `enumerateSubtrees`, which stores them in the dictionary `subtrees`, keyed by the nodes in H at which they are rooted. It does this by again making use of symmetry-breaking, and listing only one tree from each equivalence class of symmetries.

The algorithm proceeds to count the subtrees of H that appear in G ’s 1-shell and at the boundary with its 2-core. It loops through all of G ’s nodes in the same order as the k -core decomposition to quickly scan the 1-shell nodes and then remove them, making the subsequent 2-core search more tractable. We make use of the function f , which represents the partial isomorphism between a subset of H and G as the algorithm progresses in its search. For the boundary case, we associate each f with a multiplier, which simply consists of a count of the number of subtrees that “extend” from f into the 1-shell of G . `ISOMORPHICEXTENSIONS` is a

Algorithm 3.1 COUNTSUBGRAPHS(H, G)

```

//Initialization
kCoreDecompose(H)
kCoreDecompose(G)
 $ECR_H := \text{findEquivalenceClassesRepresentatives}(H)$ 
 $SBC_H := \text{findSymmetryBreakingConditions}(H)$ 
 $f := \text{initializeParitalMap}$ 

//enumerating all subtrees in  $H$ 's 1-shell
for  $h$  in  $ECR_H$  do
  if  $h \in P_H$  OR  $h$  is a root then
     $T := \text{detachTree}(H, h[, \text{parent}(h)])$ 
     $\text{subtrees}[h] := \text{enumerateSubtrees}(T, h)$ 
  end if
end for

//counting subtrees in  $G$ 's 1-shell and boundary
for  $g$  in kCoreTraversalOrder( $G$ ) do
  if  $g \in P_G$  AND  $H$  is a tree then
     $S := \text{detachTree}(G, g, \text{parent}(g))$ 
    for  $h$  in  $ECR_H$  do
      1ShellCount := COUNTROOTEDSUBTREES( $H, h, S, g$ )
    end for

  else if  $g$  is a root then
     $S := \text{detachTree}(G, g)$ 
    for  $h$  in  $ECR_H$  do
      for  $T_h$  in  $\text{subtrees}[h]$  do
        boundaryMultiplier := COUNTROOTEDSUBTREES( $T_h, h, S, g$ )
         $f.\text{setMultiplier}(\text{boundaryMultiplier})$ 
         $G.\text{markNodes}(S)$ 
        ISOMORPHICEXTENSIONS( $f, H \setminus T_h, G, SBC_H$ )
      end for
    end for
     $G.\text{removeNodes}(S)$ 
  end if
end for

//counting subgraphs in the 2-core
2CoreCount := count(FINDSUBGRAPHINSTANCES( $H, G$ ))
boundaryCount :=  $\prod_{f \in \text{boundary}} f.\text{getMultiplier}$ 

return 1ShellCount + boundaryCount + 2CoreCount

```

function introduced in [9] which incrementally extends f one vertex at a time until it maps the entirety of H onto a subset of G . By using the multiplier, we can initialize f from a root of G and let it expand inward into the 2-core, and use COUNTROOTEDSUBTREES to count partial instances of H in the 1-shell. At the end of the search, we multiply the number of instances found by ISOMORPHICEXTENSIONS by their individual multipliers to recover the total number of boundary instances. The vertices scanned by the subtree-counting routine are marked to prevent f extending into the 1-shell. After all the 1-shell vertices are taken care of, they are removed and only the 2-core of G remains.

Note that here we use a slightly modified version of ISOMORPHICEXTENSIONS that accounts for the early abort by coreness feature discussed above. Namely, when searching for an image vertex $g \in G$, ISOMORPHICEXTENSIONS first checks the coreness of g to ensure that it is greater or equal than the coreness of its tentative preimage h .

3.3. Upper bounds on runtime. The subtree-counting algorithm works by recursively computing the number of matchings at each level of a query tree T and a target tree S . To obtain an analytic upper bound on the number of steps in its execution, we count the total number of calls to COUNTROOTEDSUBTREES, and multiply that by the worst-case runtime of computing the permanent of a weighed adjacency matrix. Note that in general such a matrix will not be square, which leads to an added binomial coefficient in the permanent's runtime contribution. This is because the permanent of a non-square matrix is the product of the permanents of each of its square submatrices.

The following is an approximate expression of the number of steps in COUNTROOTEDSUBTREES' execution on trees T and S . T' is the set of non-leaf vertices of T , ϕ is the root-preserving isomorphism mapping T to a subtree of S , m_u and n_v are the number of children of vertices $u \in T$ and $v \in S$ respectively:

$$(3.2) \quad f_{crs}(T, S) \approx \sum_{u \in T'} \binom{n_{\phi(u)}}{m_u} \times m_u^2 \times 2^{m_u-1}$$

By comparison, the Grochow-Kellis algorithm will test all m vertices of T with all n vertices of S up to symmetries of T .

$$(3.3) \quad f_{gk}(T, S) \approx \frac{1}{|Aut(T)|} \binom{n}{m} = \frac{1}{|Aut(T)|} \binom{1 + \sum_{u \in T'} n_{\phi(u)}}{1 + \sum_{u \in T'} m_u}$$

In the context of counting subgraphs, COUNTROOTEDSUBTREES will be called once for each pair of non-leaf vertices in H and roots in G . To factor in the speedup afforded by the coreness condition described in Theorem 3.2, let $m = |H|$ and $n = |G|$, let m_K, n_K be the sizes of the k -cores of H and G , and R the number of roots in G . This gives an upper bound on the total runtime of the subgraph counting algorithm:

$$(3.4) \quad f_{cs}(H, G) \leq \frac{R}{|Aut(H)|} \times \sum_{T \subseteq P_H} f_{crs}(T, S) \times \binom{n_K}{m_K} \binom{n_{K-1} - m_K}{m_{K-1} - m_K} \cdots \binom{n_2 - m_3}{m_2 - m_3}$$

We compare this with an upper bound on the Grochow-Kellis algorithm’s performance:

$$(3.5) \quad f_{gk}(H, G) \leq \frac{1}{|Aut(H)|} \binom{n}{m}$$

4. Real-data experiments.

4.1. Methods. We implemented the Grochow-Kellis algorithm and [Algorithm 3.1](#) in Python 3.6 and ran them on a set of Intel Xeon E5-2680 v3 processors at 2.50GHz running Red Hat Enterprise Linux 7, from CU Boulder’s RMACC supercomputer. Because each motif/network query is a computation that can be run on a separate processor, we made use of concurrency wherever possible. The runtimes we report in the following section are expressions of the total time taken for all concurrent processes summed together.

Using data from the communityFitNet corpus [7] we tested a sample of 8 undirected networks of comparable size (around 150 vertices) spanning different topologies and categories of systems. We took care in our selection (by visual inspection) to make sure each network’s topology was typical of the networks that represent the same type of system in the communityFitNet dataset. In each network we searched for all undirected graphs up to size 8 **see figure**. Informed by the upper bounds derived in the previous section, we then related the speedup afforded by our algorithm to the following network statistics: size of the 1-shell relative to network size, maximum and average coreness of 2-core, maximum and average peripheral hub degree, and maximum and average tree depth in 1-shell. We identified **magic statistic 1** and **magic statistic 2** as the statistics most predictive of a large performance improvement in our algorithm. Namely, when **magic statistic 1** is **condition A** and **magic statistic 2** is **condition B**, our algorithm out-performs Grochow-Kellis by over a factor of C. We follow up by listing all networks from the communityFitNet corpus along with their corresponding **magic statistic 1** and **magic statistic 2**, highlighting potential applications for our algorithm. Finally, we tested four networks of around 1000 vertices whose statistics conform to the aforementioned conditions, and demonstrate our algorithm’s performance in querying all undirected graphs up to size X, as well as a sample of W graphs of sizes Y and Z.

Because our code is written in Python, it was expected that the runtimes are not comparable to those of available motif-finding software, which is often written in faster languages such as Java [9] or C++ [24] and optimized for experimental use. Our aim was to provide a common-platform comparison with a standard motif-centric algorithm from the literature.

4.2. Results and discussion. Because our code is written in Python, it was expected that the runtimes are not comparable to those of available motif-finding software, which is often written in faster languages such as Java [9] or C++ [24] and optimized for experimental use. Our aim was to provide a common-platform comparison with a standard motif-centric algorithm from the literature. Because our code is written in Python, it was expected that the runtimes are not comparable to those of available motif-finding software, which is often written in faster languages such as Java [9] or C++ [24] and optimized for experimental use. Our aim was to provide a common-platform comparison with a standard motif-centric algorithm from the literature. Because our code is written in Python, it was expected that the runtimes are not comparable to those of available motif-finding software, which is often written in faster

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323 to provide a common-platform comparison with a standard motif-centric algorithm from the
324 literature.

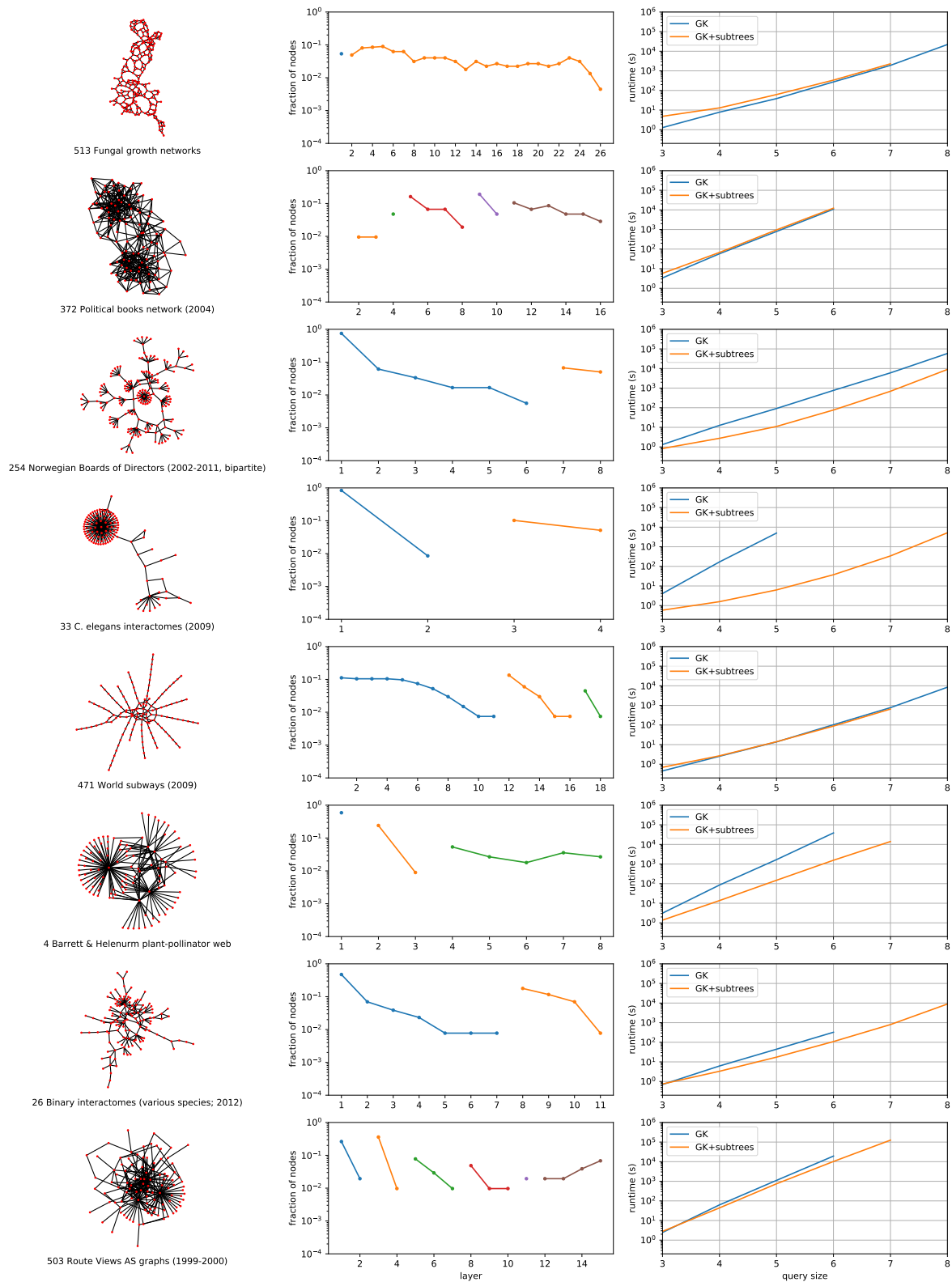


Figure 2.

5. Conclusion.

REFERENCES

- [1] I. ALBERT AND R. ALBERT, *Conserved network motifs allow protein-protein interaction prediction*, Bioinformatics, 20 (2004), pp. 3346–3352.
- [2] N. ALON, P. DAO, I. HAJIRASOULIHA, F. HORMOZDIARI, AND S. C. SAHINALP, *Biomolecular network motif counting and discovery by color coding*, Bioinformatics, 24 (2008), pp. i241–i249.
- [3] A.-L. BARABÁSI, E. RAVASZ, AND T. VICSEK, *Deterministic scale-free networks*, Physica A: Statistical Mechanics and its Applications, 299 (2001), pp. 559–564.
- [4] A. D. BROIDO AND A. CLAUSET, *Scale-free networks are rare*, Nature communications, 10 (2019), pp. 1–10.
- [5] R. CURTICAPEAN AND D. MARX, *Complexity of counting subgraphs: Only the boundedness of the vertex-cover number counts*, in 2014 IEEE 55th Annual Symposium on Foundations of Computer Science, IEEE, 2014, pp. 130–139.
- [6] F. V. FOMIN, D. LOKSHTANOV, V. RAMAN, S. SAURABH, AND B. R. RAO, *Faster algorithms for finding and counting subgraphs*, Journal of Computer and System Sciences, 78 (2012), pp. 698–706.
- [7] A. GHASEMIAN, H. HOSSEINMARDI, AND A. CLAUSET, *Evaluating and comparing overfit in models of network community structure*, Preprint, (2017).
- [8] L. A. GOLDBERG AND M. JERRUM, *Counting unlabelled subtrees of a tree is $\#P$ -complete*, LMS Journal of Computation and Mathematics, 3 (2000), pp. 117–124.
- [9] J. A. GROCHOW AND M. KELLIS, *Network motif discovery using subgraph enumeration and symmetry-breaking*, in Annual International Conference on Research in Computational Molecular Biology, Springer, 2007, pp. 92–106.
- [10] S. ITZKOVITZ, R. LEVITT, N. KASHTAN, R. MILO, M. ITZKOVITZ, AND U. ALON, *Coarse-graining and self-dissimilarity of complex networks*, Physical Review E, 71 (2005), p. 016127.
- [11] S. ITZKOVITZ, R. MILO, N. KASHTAN, G. ZIV, AND U. ALON, *Subgraphs in random networks*, Physical review E, 68 (2003), p. 026127.
- [12] Z. R. M. KASHANI, H. AHRABIAN, E. ELAHI, A. NOWZARI-DALINI, E. S. ANSARI, S. ASADI, S. MOHAMMADI, F. SCHREIBER, AND A. MASOUDI-NEJAD, *Kavosh: a new algorithm for finding network motifs*, BMC bioinformatics, 10 (2009), pp. 1–12.
- [13] N. KASHTAN, S. ITZKOVITZ, R. MILO, AND U. ALON, *Efficient sampling algorithm for estimating subgraph concentrations and detecting network motifs*, Bioinformatics, 20 (2004), pp. 1746–1758.
- [14] Y.-X. KONG, G.-Y. SHI, R.-J. WU, AND Y.-C. ZHANG, *K-core: Theories and applications*, Physics Reports, 832 (2019), pp. 1–32.
- [15] G. LI, J. LUO, Z. XIAO, AND C. LIANG, *Mtmo: an efficient network-centric algorithm for subtree counting and enumeration*, Quantitative Biology, 6 (2018), pp. 142–154.
- [16] W. LIN, X. XIAO, X. XIE, AND X.-L. LI, *Network motif discovery: A gpu approach*, IEEE transactions on knowledge and data engineering, 29 (2016), pp. 513–528.
- [17] R. MILO, S. SHEN-ORR, S. ITZKOVITZ, N. KASHTAN, D. CHKLOVSKII, AND U. ALON, *Network motifs: simple building blocks of complex networks*, Science, 298 (2002), pp. 824–827.
- [18] M. NEWMAN, *Networks*, Oxford university press, 2018.
- [19] T. OHNISHI, H. TAKAYASU, AND M. TAKAYASU, *Network motifs in an inter-firm network*, Journal of Economic Interaction and Coordination, 5 (2010), pp. 171–180.
- [20] S. OMIDI, F. SCHREIBER, AND A. MASOUDI-NEJAD, *Moda: an efficient algorithm for network motif discovery in biological networks*, Genes & genetic systems, 84 (2009), pp. 385–395.
- [21] P. V. PAULAU, C. FEENDERS, AND B. BLASIUS, *Motif analysis in directed ordered networks and applications to food webs*, Scientific reports, 5 (2015), p. 11926.
- [22] N. PRŽULJ, D. G. CORNEIL, AND I. JURISICA, *Modeling interactome: scale-free or geometric?*, Bioinformatics, 20 (2004), pp. 3508–3515.
- [23] P. RIBEIRO AND F. SILVA, *G-tries: an efficient data structure for discovering network motifs*, in Proceedings of the 2010 ACM symposium on applied computing, 2010, pp. 1559–1566.
- [24] P. RIBEIRO AND F. SILVA, *G-tries: a data structure for storing and finding subgraphs*, Data Mining and Knowledge Discovery, 28 (2014), pp. 337–377.

- 377 [25] H. J. RYSER, *Combinatorial mathematics*, vol. 14, American Mathematical Soc., 1963.
- 378 [26] Y. SHAO, B. CUI, L. CHEN, L. MA, J. YAO, AND N. XU, *Parallel subgraph listing in a large-scale graph*,
379 in Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data, 2014,
380 pp. 625–636.
- 381 [27] S. S. SHEN-ORR, R. MILO, S. MANGAN, AND U. ALON, *Network motifs in the transcriptional regulation*
382 *network of escherichia coli*, Nature genetics, 31 (2002), pp. 64–68.
- 383 [28] O. SPORNS AND R. KÖTTER, *Motifs in brain networks*, PLoS Biol, 2 (2004), p. e369.
- 384 [29] S. STROGATZ, *Exploring complex networks*, nature, 410 (2001), pp. 268–276.
- 385 [30] L. G. VALIANT, *The complexity of computing the permanent*, Theoretical computer science, 8 (1979),
386 pp. 189–201.
- 387 [31] L. G. VALIANT, *The complexity of enumeration and reliability problems*, SIAM Journal on Computing, 8
388 (1979), pp. 410–421.
- 389 [32] S. VALVERDE AND R. V. SOLÉ, *Network motifs in computational graphs: A case study in software*
390 *architecture*, Physical Review E, 72 (2005), p. 026107.
- 391 [33] S. WERNICKE AND F. RASCHE, *Fanmod: a tool for fast network motif detection*, Bioinformatics, 22
392 (2006), pp. 1152–1153.